K M Vinayak Ram

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EDUCATION

MS Ramaiah University Of Applied Sciences || BTech in Computer Science || **2023-2027**

Coursework: Python Programming, JAVA (Basics), Data Base Management System, Data Analytics, Data Structure and Algorithm, System Architecture.

TECHNICAL SKILLS

- **Programming:** Python, C, Java, MySQL
- **Tools and libraries:** OpenCV, NumPy, Jupyter Notebooks, Git, React.js
- **Other Skills**: Object-Oriented Programming, Version Control, Debugging, Code Optimization
- **Web development**: React.js, Node.js, Express, MongoDB, HTML, CSS, , JavaScript

EXPERIENCE

Toyota Tsusho India **Bangalore, India**

Software Engineering Intern *September 2026*

Tools: Java Script, React, Microsoft Azure

- **Leading the development of a digital accident reporting solution** that streamlines incident reporting by enabling drivers to submit accident details and upload photos online.
- **Implementing role-based dashboards** for drivers, location-specific admins, and super admins, enabling local administrators to manage reports from drivers within their respective locations and providing super admins with nationwide visibility into accident reports across India.
- **Managing deployment on Microsoft Azure** to host and launch the application in a live environment.

Indus Engineering **Bangalore, India**

Full Stack Web Development Internship *July 2025 - Sep2025*

Tools: Java Script, React, Python, Figma, NewsOrg API

- **Architected responsive and user-centric web interfaces** using Figma for prototyping and React.js for front-end implementation, ensuring intuitive navigation, cross-device compatibility, and a seamless user experience.
- **Developed a custom web scraping engine** in Python to extract data based on dynamic user-defined keywords, improving data accessibility and reducing manual information retrieval efforts. Integrated Python-based back-end services with a React.js front-end using RESTful API principles, enabling efficient communication, real-time search results, and smooth data exchange between the application layers.
- **Utilized the NewsOrg API** to fetch, process, and dynamically display live news data feeds, ensuring users received up-to-date information through an interactive and continuously refreshed interface.

Innovate Intern **Bangalore, India**

Python Programming and Development Internship *July 2024 - Aug2024*

Tools: Python, OpenCV, Numpy

- **Worked on a guided human detection and crowd-counting project** using Python and OpenCV, gaining hands-on experience with basic computer vision and real-time video processing. Explored image processing techniques such as frame-by-frame comparison and background subtraction to understand how motion and human activity can be detected from video streams.
- **Applied NumPy for numerical and array-based processing** while working with live video data, improving understanding of efficient image and video manipulation.
- **Used Git for basic version control and Jupyter Notebooks** for experimentation, testing, and documenting different computer vision concepts during the project.

PROJECTS

Employee Attrition Prediction & Model Interpretability

- Developed an end-to-end machine learning classification pipeline using **Random Forest** to forecast employee churn and identify key drivers of turnover.
- Handled severe dataset class imbalance using **SMOTE (Synthetic Minority Over-sampling Technique)** to enhance model sensitivity and prediction reliability.
- Integrated **SHAP (SHapley Additive exPlanations)** to translate complex model outputs into transparent, business-friendly insights, aligning technical results with actionable HR retention strategies.

NotesFlow | Online Notes Sharing Application

- Designed & developed a notes sharing web application that enables users to upload and access academic notes through an intuitive and user friendly interface.
- Integrated AI-powered functionalities to improve intelligent search, enhancing the efficiency of note management and user interaction within the platform. Utilized modern frontend and backend technologies along with database integration to ensure secure data handling, seamless performance, and scalable application architecture.
- Ensured cross-platform compatibility and optimized the application for smooth performance across desktops and mobile devices by applying responsive design principles using CSS techniques such as Flexbox and Grid.

AmbuBeam | AI-Powered Green Corridor System

- Conceptualized an automated traffic override system designed to eliminate ambulance delays at intersections using AI and IoT.
- Integrated audio sensors and smart reflectors to detect specific ambulance siren frequencies, triggering immediate signal changes.
- Leveraged RF/LoRa modules for robust long-range wireless communication and implemented solar-powered energy solutions for sustainability.
- Developed signal override logic to prioritize emergency vehicles while maintaining overall traffic flow safety.

CERTIFICATIONS

- **CISCO NETWORKING ACADEMY-** Introduction to Data Science
  Completed coursework covering the data science lifecycle, data collection and cleaning, exploratory data analysis, and basic machine learning using Python. Developed understanding of key libraries such as Pandas, NumPy, and Matplotlib.
- **IBM COGNITIVE CLASSES-** Python for Data Science
  Completed hands-on coursework covering Python fundamentals, data manipulation with Pandas and NumPy, data visualization with Matplotlib, and introduction to statistical and machine learning concepts.